

Multi-Sided Platform Architecture—a digital business model where a platform connects two or more different groups of users and creates value by helping them interact.

A simple way to understand it is:

The platform itself may not provide the main product or service. Instead, it creates the environment where different groups can find, communicate, transact, and exchange value.

For example, **Uber does not need to own every car.** It connects **riders who need transportation** with **drivers who provide transportation.**

1. Two-Sided Market Mechanics

What does it mean?

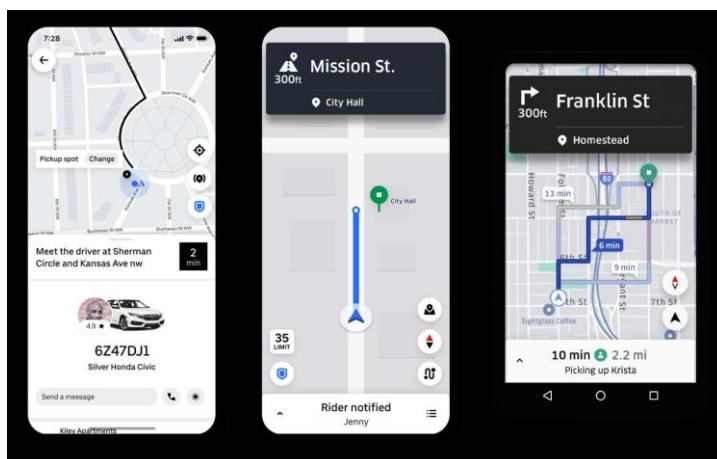
A **two-sided market** connects two distinct groups that need each other.

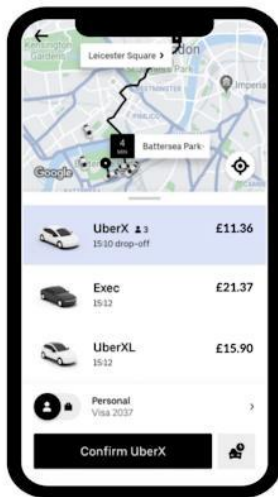
Typical combinations include:

- **Buyers ↔ Sellers**
- **Drivers ↔ Riders**
- **Guests ↔ Hosts**
- **Students ↔ Tutors**
- **Freelancers ↔ Clients**
- **Restaurants ↔ Customers**

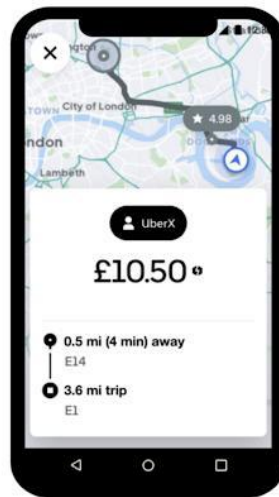
The platform acts as the **intermediary** that makes the interaction easier.

Example: Uber





Rider App



Driver App



Side 1 – Riders

- Need transportation.
- Open Uber.
- Enter their destination.
- Request a ride.

Side 2 – Drivers

- Want passengers and income.
- Use the Uber Driver app.
- Receive ride requests.
- Pick up passengers.

Uber – Platform

- Matches riders with drivers.
- Calculates fares.
- Processes payments.
- Provides GPS/navigation.
- Collects ratings.
- Handles communication.

The value exchange

Rider → Uber → Driver

Rider needs a ride → Uber finds a driver → Driver provides the ride → Rider pays.

So Uber's primary role is **connecting the two sides efficiently**.

2. Network Effects

What does it mean?

A **network effect** occurs when a platform becomes more valuable as more users participate.

There are two important types.

A. Cross-Side Network Effect

One group becoming larger makes the platform more valuable to another group.

Uber example

Imagine Uber has:

100 drivers + 1,000 riders

Drivers have many potential customers.

Now Uber grows to:

10,000 drivers + 500,000 riders

More riders create more opportunities for drivers to earn money.

At the same time, more drivers mean riders can potentially get:

- Faster matching
- More available vehicles

- Better geographic coverage
- Potentially shorter waiting times

So:

More drivers → more value for riders

More riders → more value for drivers

This is a **cross-side network effect**.

B. Same-Side Network Effect

The number of users on the **same side** affects the value experienced by other users on that side.

For example, consider **LinkedIn**.

More professionals joining LinkedIn can create:

More professionals → more professional connections → more networking opportunities

Similarly:

More employers → more job opportunities → more value for job seekers

However, same-side effects are not always positive. For example, if too many sellers compete on the same marketplace, individual sellers may face more competition.

3. Platform Orchestration

What does it mean?

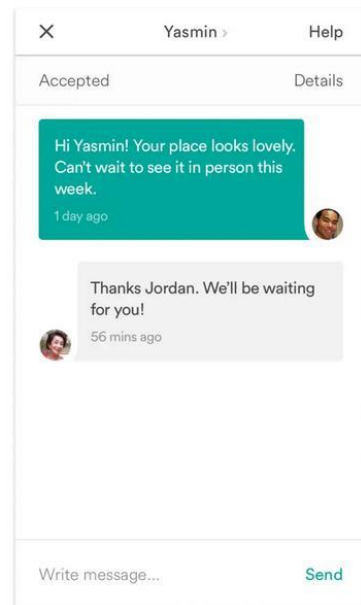
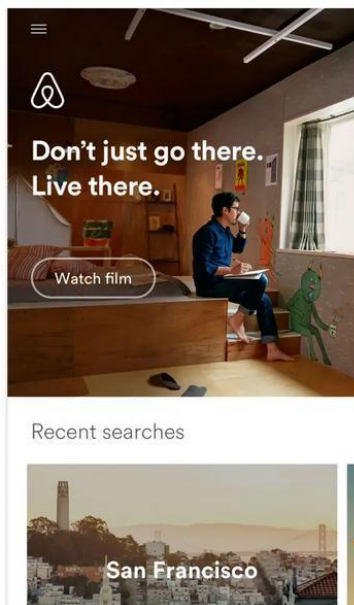
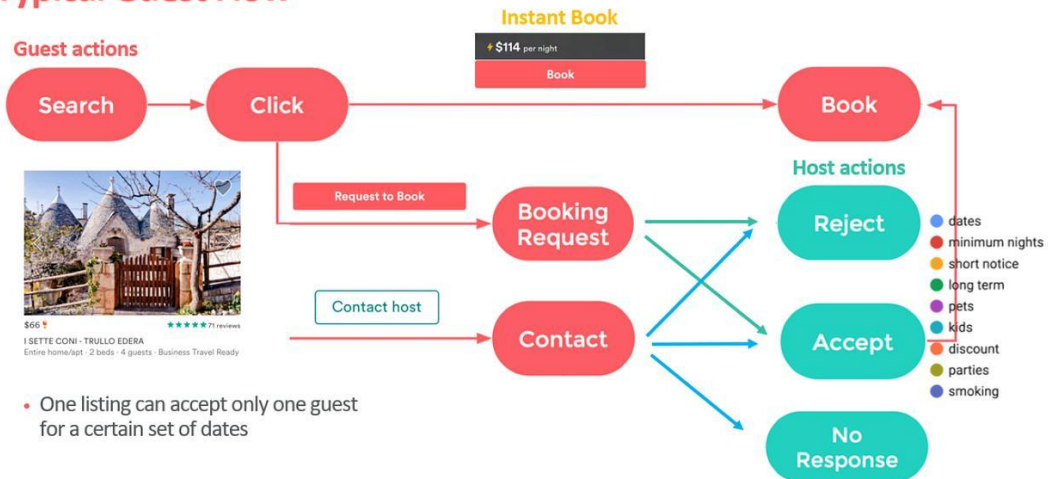
Platform orchestration means managing and coordinating all the activities that allow the different sides of the platform to interact successfully.

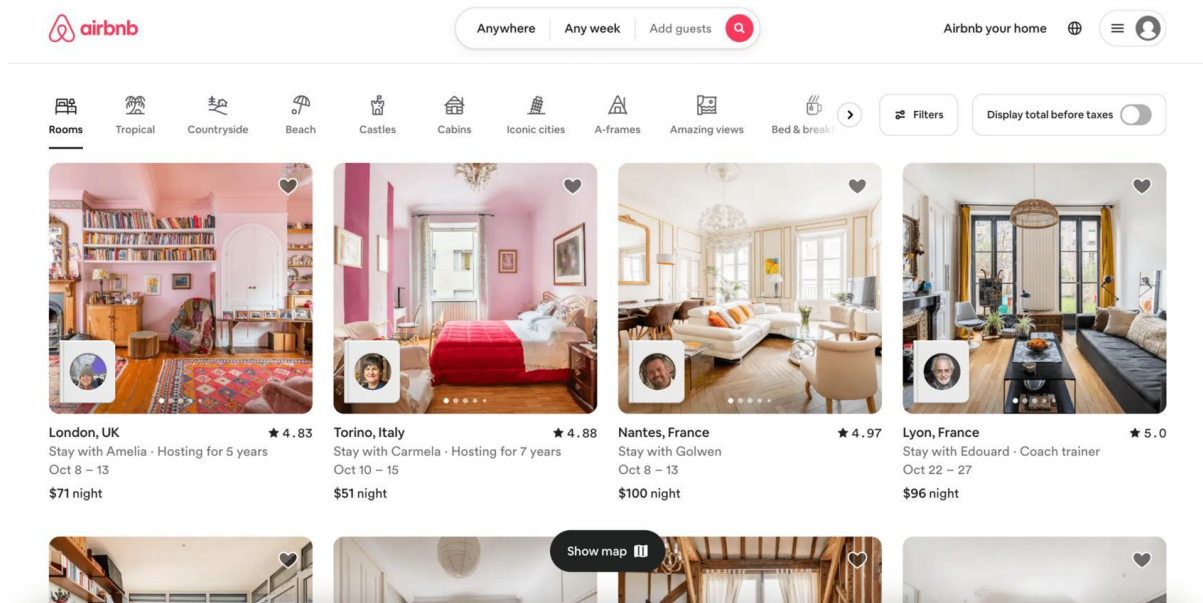
Think of the platform as a **digital conductor of an orchestra**.

It does not necessarily perform every activity itself. Instead, it coordinates everyone.

Example: Airbnb

Typical Guest Flow





Airbnb connects:

Hosts ↔ Airbnb ↔ Guests

Airbnb has to coordinate many things:

Activity	How Airbnb helps
Matching	Guests search for suitable accommodation
Pricing	Hosts can use pricing tools
Booking	Guests make reservations
Payment	Airbnb facilitates payments
Reviews	Guests and hosts review each other
Trust	Identity verification and platform policies
Communication	Messaging between host and guest
Dispute handling	Platform processes certain complaints
Recommendations	Algorithms suggest properties

Without this orchestration, the marketplace would become difficult to use.

Example

Suppose a guest searches:

"3-bedroom apartment in Singapore for 4 people, 5 nights."

Airbnb's platform can filter and rank properties based on:

- Location
- Price
- Availability
- Number of bedrooms
- Reviews
- Amenities
- Guest preferences

The platform is therefore **orchestrating the interaction** between the guest and host.

4. Match-Making Algorithms

This is an important part of platform orchestration.

A platform uses algorithms to determine:

Who should be connected with whom?

Uber example

A rider requests a trip from:

Orchard Road → Changi Airport

Uber's system considers factors such as:

- Driver location
- Driver availability
- Destination
- Estimated travel time
- Traffic conditions
- Ride type

The system then identifies an appropriate driver.

So:

Rider request → Algorithm → Suitable driver

The algorithm reduces the amount of time users would otherwise spend searching manually.

5. Automated Pricing

Platforms can also use technology to adjust prices or provide pricing recommendations.

Example: Airbnb

Suppose there is a major event in Singapore.

Normally:

Hotel/room demand = moderate

During the event:

Hotel/room demand = very high

Hosts may increase their prices because demand has increased.

Similarly, platforms can use data about:

- Demand
- Availability
- Location
- Season
- Historical bookings
- Events

to support pricing decisions.

Uber example

Uber has also used **dynamic pricing**, where fares can change based on factors such as demand and available drivers.

For example:

Normal demand → normal fare

Very high demand + limited drivers → higher fare

The important concept for students is:

Digital platforms can use data and algorithms to coordinate supply and demand.

6. Digital Trust Mechanisms

One major problem with online marketplaces is:

"How do I trust someone I don't know?"

Platforms solve this using various trust mechanisms.

Example: Airbnb

A guest may think:

"How do I know this host is legitimate?"

Airbnb can provide mechanisms such as:

- User profiles
- Reviews
- Ratings
- Identity-related verification
- Booking records
- Payment protection mechanisms
- Platform policies

Similarly, a host may ask:

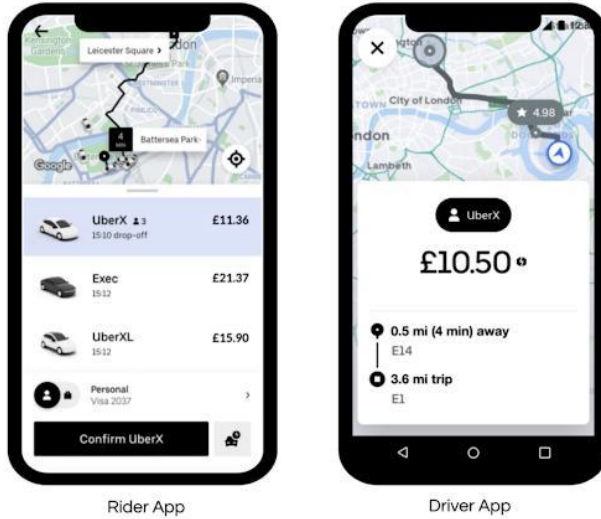
"How do I know this guest is trustworthy?"

The platform's reputation and review system helps reduce uncertainty.

7. Market Champions

The slide gives three excellent examples.

A. Uber – Mobility



Platform sides

Drivers ↔ Uber ↔ Riders

Value created

Drivers get:

- Access to customers
- Income opportunities
- Digital tools

Riders get:

- Access to transportation
- Driver matching
- Digital payment
- Trip tracking

Platform advantage

Uber's value comes largely from **connecting and coordinating** the two sides.

8. Airbnb – Hospitality



Absolute Oceanfront Beach House

4.81 (34 reviews)

Superhost

Pahoa, Hawaii, United States

Share

Save







Show all photos

Entire house hosted by Dan

2 guests · 1 bedroom · 1 bed · 1 bath

Entire home

You'll have the house to yourself.

Enhanced Clean

This host committed to Airbnb's 5-step enhanced cleaning process. [Learn more](#)

Self check-in

Check yourself in with the keypad.

House rules

This place isn't suitable for children under 12 and the host doesn't allow pets, parties, or smoking. [Get details](#)

Free cancellation for 48 hours

\$265 / night

4.81 (34 reviews)

CHECK IN

7/25/2021

CHECK OUT

8/1/2021

GUESTS

2 guests

Reserve

You won't be charged yet

\$279 x 7 nights

\$1,954

5% weekly price discount

-\$98

Cleaning fee

\$200

Service fee

\$290

Occupancy taxes and fees

\$308

Total

\$2,654

Report this listing

The Beach House sits right on the cliff with amazing views the ocean, Kehena is only a 5 min walk. Located on the Puna coast with many black sand beaches, and away from it all. You have 4 amazing locals with ocean views to choose from. Very private, gated and surrounded by lush gardens! Decorated it in Hawaiiana style, it is located on a dead end street, the neighborhood is quiet and the Puna coast offers many w... [read more](#)

airbnb

Settings

Become a host

SIGN UP OR LOGIN

Book your dream vacation

Where would you like to stay?

Guests

3/28/2022

3/28/2022

Q

Platform sides

Hosts ↔ Airbnb ↔ Guests

Hosts provide:

- Houses
- Apartments
- Rooms
- Other accommodation

Guests provide:

- Demand
- Payments

- Reviews

Airbnb provides:

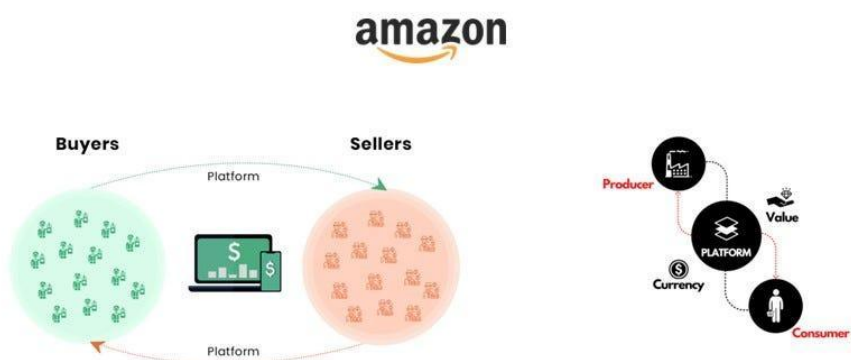
- Search
- Matching
- Booking
- Payment
- Communication
- Reviews
- Trust mechanisms

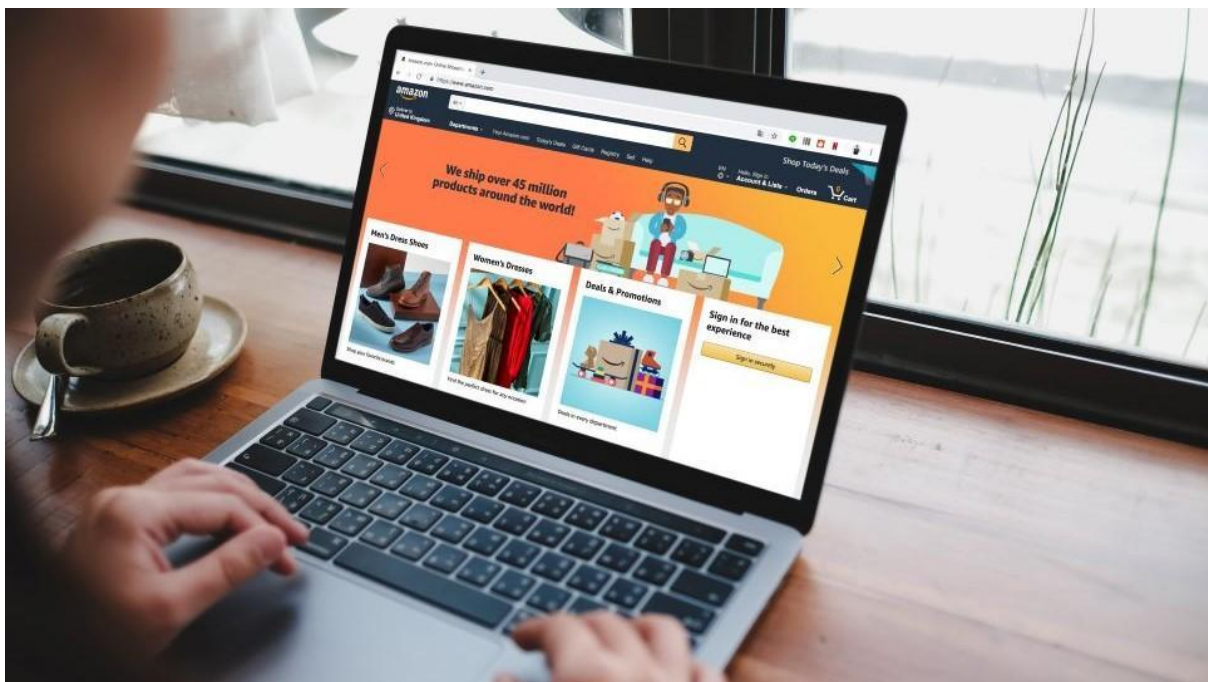
Therefore:

Airbnb creates a marketplace rather than owning all the accommodation listed on its platform.

9. Amazon Marketplace – E-Commerce

Market place Business Model





Amazon Marketplace connects:

Sellers ↔ Amazon ↔ Buyers

Seller side

Businesses and individuals can list products.

Buyer side

Customers search for and purchase products.

Amazon's role

Amazon provides infrastructure such as:

- Product listings
- Search
- Recommendations
- Payment
- Reviews
- Order management
- Logistics/fulfilment options
- Seller tools

Network effect

More sellers can mean:

More products → more choices for customers

More customers can mean:

More potential buyers → more opportunities for sellers

This can create a reinforcing cycle:

More sellers → more products → more customers → more sales opportunities → more sellers

10. Feedback Loops and Data Flow

This is another important part of the diagram.

Every interaction on a digital platform generates **data**.

For example, on an e-commerce platform:

Customer searches for running shoes

↓

Views products

↓

Adds one to cart

↓

Purchases

↓

Gives a rating

↓

Platform collects data

↓

Algorithm improves recommendations

↓

Customer receives more relevant products

This creates a **feedback loop**.

Example: Netflix

A simplified version is:

Watch movie → Netflix learns preferences → recommends similar movie → user watches → more data → better recommendations

The platform becomes better at personalization as it collects more interaction data.

11. Value Creation vs. Value Capture

This distinction is particularly useful for students studying digital business.

Value Creation

The platform creates value by helping users do something more easily.

For example, Amazon helps:

Buyer find seller + Seller find buyer

Value Capture

The platform generates revenue from the activity.

For example, a marketplace may generate revenue through:

- Transaction fees
- Commissions
- Advertising
- Subscription fees
- Premium services

So:

Value Creation = How the platform creates benefits for users

Value Capture = How the platform earns money from those activities

12. Putting Everything Together: Uber Example

Step 1 – Two sides

Drivers ↔ Riders

Step 2 – Platform

Uber provides the digital infrastructure connecting them.

Step 3 – Matchmaking

The algorithm identifies an available driver for a rider.

Step 4 – Transaction

The rider books the ride and pays digitally.

Step 5 – Trust

Both sides can rate each other.

Step 6 – Data

Uber collects information about:

- Trips
- Locations
- Demand
- Driver availability
- Ratings
- User behaviour

Step 7 – Feedback

The data can be used to improve:

- Matching
- Estimated arrival times
- Recommendations
- Pricing
- Platform operations

Step 8 – Network effect

More riders can attract more drivers.

More drivers can improve availability for riders.

Step 9 – Revenue

The platform captures value through its business model, including fees associated with transactions.

Concept	Simple Meaning	Real-World Example
Two-Sided Market	Connect two groups that need each other	Uber: drivers ↔ riders
Network Effect	More users can increase platform value	More Uber drivers → better availability
Platform Orchestration	Coordinate interactions between users	Airbnb manages search, booking, payment and reviews
Matchmaking	Algorithm connects suitable users	Uber matches riders with drivers
Automated Pricing	Technology supports dynamic/recommended prices	Ride/accommodation pricing based on demand
Digital Trust	Mechanisms reduce uncertainty between strangers	Ratings, reviews, verification
Feedback Loop	User activity generates data that improves the platform	Amazon recommendations
Value Creation	Benefits created for platform participants	Amazon connects buyers and sellers
Value Capture	How the platform makes money	Commission, transaction fees, advertising
Market Champion	Major real-world platform example	Uber, Airbnb, Amazon